



ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTTP/ELECT MAINT/2024-25/LMIRNGFL/452310

Date:

29/08/2024(18:04:56)

Subject:LMI on RING FLUX TESTING OF GENERATOR STATOR
CORES

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes were suggested by COS.

Put up for Periodical reviewing of the LMI.

May please approve.

Submitted by:

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On:09/09/2024(18:43:19) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIRNGFL/452310

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**RING FLUX TESTING OF GENERATOR STATOR
CORES**

LMI/OIN/OPS/ELEC/012

Aug 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	RING FLUX TESTING OF GENERATOR STATOR CORES
2	LMI No.	LMI/OIN/OPS/ELEC/012 Rev 00
3	Reference Document	LMI/OIN/OPS/ELEC/012
4	Superseded Document	NIL
5	Prepared By	Name: Balabalaji Akula DGM (EMD)
6	Checked By	Name: Nageswara rao, DGM (EMD)
7	Reviewed By	Name: T Vijaykrishna AGM (EM)-I/C
8	Reviewed By	Name: S.J.David GM (O&M)
	Approved By	Name: S.K.Sinha HOP (Simhadri)

TITLE	RING FLUX TESTING OF GENERATOR STATOR CORES
LMI No.	LMI/OIN/OPS/ELEC/012 Rev 00

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TITLE	RING FLUX TESTING OF GENERATOR STATOR CORES
LMI No.	LMI/OIN/OPS/ELEC/012 Rev 00

1.0 Introduction

Generator stator inter lamination insulation may be damaged during assembly or maintenance and also during removal and replacement of its rotor. Hot spots may also develop during operation due to ingress of foreign bodies into the bore or due to deterioration of inter lamination insulation. These should therefore be detected and repaired during Generator major overhauls.

2.0 Superseded Documents

NIL

3.0 Details of Revision

NIL

4.0 METHODOLOGY OF IMPLEMENTATION

Low Voltage Ring Flux Test (EL-CID Test - Electromagnetic Core imperfection Detector Test):

This employs the modern technique of detecting the core faults. In this test, the core is excited by much lower power, which gives about 4% of the rated flux in the core. The voltage gradient axially along the core induces currents to flow through any damaged areas but because the current is small therefore the heating is insignificant. The test relies on the electromagnetic detection of these fault currents following axially through the damaged region "This test enables to detect deep seated core faults in the generator core." (Covered in LMI/OGN/OPS/ELEC /014).

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H.V. Ring Flux Test.

The excitation cables are wound through the bore and over the core back. They must be arranged so that a clear view of the bore surface can be obtained. The cables may be supported by spacers hold the cables away from the bore surface, or they may be wound loosely in a group so that they can be moved away from an obscured area.

Since the excitation requirements rise very rapidly with voltage it has been decided that achieving 80 per cent of the rated core flux is acceptable the excitation should normally be taken from the station bus bars after adapting the relevant instrumentation and protection systems for the duration of the test.

The criterion used for detecting damage is that the temperature difference between a hot spot and an equivalent normal area should not exceed 10oC after one hour at an excitation of 80 percent of the rated flux. If these limits exceeds, repair work should be carried out. After all repair work the same 10o limits should be adopted. If this limit cannot be achieved advice should be sought from manufacturer.

A repeat Low Voltage ELCID test is recommended to compare with the initial test values, which will give an indication of any changes in the core brought about by the HV ring flux test.

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5.0 FREQUENCY OF CHECKING AND MEASUREMENTS:

The following is a broader list of the possible occasions when a ring flux test may be desirable.

- i) Generator has been completely rebuilt; ring flux tests should be carried out before and after winding.
- ii) When visible damage is present and repair work is carried out.
- iii) When the ELCID test indicates any abnormality.

The decision on whether to test a generator under above conditions will depend on previous experience of the generator concerned and on operational constraints.

6.0 The RESPONSIBILITY CENTERS FOR IMPLEMENTATION

AGM / DGM. of concerned area.

7.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION

Exception will be reported to HOD (EM) and can be dispensed with Approval.

8.0 Review

HOD (EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

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9.0 Checklist:**CHECKLIST DURING OVERHAUL**

SLOT NO.	LEAKAGE CURRENT (mA)	SLOT NO.	LEAKAGE CURRENT (mA)	SLOT NO.	LEAKAGE CURRENT (mA)
1		17		33	
2		18		34	
3		19		35	
4		20		36	
5		21		37	
6		22		38	
7		23		39	
8		24		40	
9		25		41	
10		26		42	
11		27		43	
12		28		44	
13		29		45	
14		30		46	
15		31		47	
16		32		48	

REMARKS:

Leakage current values are found less than 100 mA.